

5. A raindrop falls from rest from a cloud. The velocity, $v \text{ ms}^{-1}$ vertically downwards, of the raindrop, t seconds after the raindrop starts to fall, is modelled by the differential equation

$$(t+2)\frac{dv}{dt} + 3v = k(t+2) - 3 \quad t \geq 0$$

where k is a positive constant.

- (a) Solve the differential equation to show that

$$v = \frac{k}{4}(t+2) - 1 + \frac{4(2-k)}{(t+2)^3} \quad (5)$$

Given that $v = 4$ when $t = 2$

- (b) determine, according to the model, the velocity of the raindrop 5 seconds after it starts to fall.

- (c) Comment on the validity of the model for very large values of t



8. A scientist is studying the effect of introducing a population of type A bacteria into a population of type B bacteria.

At time t days, the number of type A bacteria, x , and the number of type B bacteria, y , are modelled by the differential equations

$$\frac{dx}{dt} = x + y$$

$$\frac{dy}{dt} = 3y - 2x$$

- (a) Show that

$$\frac{d^2x}{dt^2} - 4\frac{dx}{dt} + 5x = 0 \quad (3)$$

- (b) Determine a general solution for the number of type A bacteria at time t days. (4)
- (c) Determine a general solution for the number of type B bacteria at time t days. (2)

The model predicts that, at time T hours, the number of bacteria in the two populations will be equal.

Given that $x = 100$ and $y = 275$ when $t = 0$

- (d) determine the value of T , giving your answer to 2 decimal places. **(5)**
- (e) Suggest a limitation of the model. **(1)**



5. A tank at a chemical plant has a capacity of 250 litres. The tank initially contains 100 litres of pure water.

Salt water enters the tank at a rate of 3 litres every minute. Each litre of salt water entering the tank contains 1 gram of salt.

It is assumed that the salt water mixes instantly with the contents of the tank upon entry.

At the instant when the salt water begins to enter the tank, a valve is opened at the bottom of the tank and the solution in the tank flows out at a rate of 2 litres per minute.

Given that there are S grams of salt in the tank after t minutes,

- (a) show that the situation can be modelled by the differential equation

$$\frac{dS}{dt} = 3 - \frac{2S}{100 + t} \quad (4)$$

- (b) Hence find the number of grams of salt in the tank after 10 minutes.

(5)

When the concentration of salt in the tank reaches 0.9 grams per litre, the valve at the bottom of the tank must be closed.

- (c) Find, to the nearest minute, when the valve would need to be closed.

(3)

- (d) Evaluate the model.

(1)

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- $$\begin{aligned}\frac{dw}{dt} &= \frac{5}{2}(w - s) \\ \frac{ds}{dt} &= \frac{2}{5}w - 90e^{-t}\end{aligned}$$

- (e) Suggest a limitation of the model. (1)



7. A sample of bacteria in a sealed container is being studied.

The number of bacteria, P , in thousands, is modelled by the differential equation

$$(1+t)\frac{dP}{dt} + P = t^{\frac{1}{2}}(1+t)$$

where t is the time in hours after the start of the study.

Initially, there are exactly 5000 bacteria in the container.

- (a) Determine, according to the model, the number of bacteria in the container 8 hours after the start of the study.

(6)

- (b) Find, according to the model, the rate of change of the number of bacteria in the container 4 hours after the start of the study.

(4)

- (c) State a limitation of the model.

(1)

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3. (a) Determine the general solution of the differential equation

$$\cos x \frac{dy}{dx} + y \sin x = e^{2x} \cos^2 x$$

giving your answer in the form $y = f(x)$

(3)

Given that $y = 3$ when $x = 0$

- (b) determine the smallest positive value of x for which $y = 0$

(3)

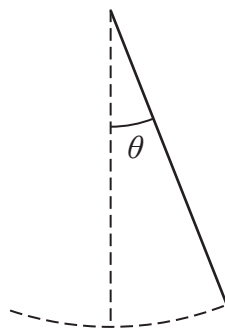
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10.

**Figure 3**

The motion of a pendulum, shown in Figure 3, is modelled by the differential equation

$$\frac{d^2\theta}{dt^2} + 9\theta = \frac{1}{2}\cos 3t$$

where θ is the angle, in radians, that the pendulum makes with the downward vertical, t seconds after it begins to move.

(a) (i) Show that a particular solution of the differential equation is

$$\theta = \frac{1}{12}t \sin 3t \quad (4)$$

(ii) Hence, find the general solution of the differential equation. (4)

Initially, the pendulum

- makes an angle of $\frac{\pi}{3}$ radians with the downward vertical
- is at rest

Given that, 10 seconds after it begins to move, the pendulum makes an angle of α radians with the downward vertical,

(b) determine, according to the model, the value of α to 3 significant figures. (4)

Given that the true value of α is 0.62

(c) evaluate the model. (1)

The differential equation

$$\frac{d^2\theta}{dt^2} + 9\theta = \frac{1}{2}\cos 3t$$

models the motion of the pendulum as moving with forced harmonic motion.

(d) Refine the differential equation so that the motion of the pendulum is simple harmonic motion. (1)

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6. The motion of a particle P along the x -axis is modelled by the differential equation

$$2\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 2x = 4t + 12$$

where P is x metres from the origin O at time t seconds, $t \geq 0$

- (a) Determine the general solution of the differential equation. (6)

- (b) Hence determine the particular solution for which $x = 3$ and $\frac{dx}{dt} = -2$ when $t = 0$
- (3)**

- (c) (i) Show that, according to the model, the minimum distance between O and P is $(2 + \ln 2)$ metres.
- (ii) Justify that this distance is a minimum.
- (4)**

For large values of t the particle is expected to move with constant speed.

- (d) Comment on the suitability of the model in light of this information. (1)



5. An engineer is investigating the motion of a sprung diving board at a swimming pool. Let E be the position of the end of the diving board when it is at rest in its equilibrium position and when there is no diver standing on the diving board. A diver jumps from the diving board. The vertical displacement, h cm, of the end of the diving board above E is modelled by the differential equation

$$4 \frac{d^2h}{dt^2} + 4 \frac{dh}{dt} + 37h = 0$$

where t seconds is the time after the diver jumps.

- (a) Find a general solution of the differential equation.

(2)

When $t = 0$, the end of the diving board is 20 cm below E and is moving upwards with a speed of 55 cm s^{-1} .

- (b) Find, according to the model, the maximum vertical displacement of the end of the diving board above E .

(8)

- (c) Comment on the suitability of the model for large values of t .

(2)

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3. A scientist is investigating the concentration of antibodies in the bloodstream of a patient following a vaccination.

The concentration of antibodies, x , measured in micrograms (μg) per millilitre (ml) of blood, is modelled by the differential equation

$$100 \frac{d^2x}{dt^2} + 60 \frac{dx}{dt} + 13x = 26$$

where t is the number of weeks since the vaccination was given.

- (a) Find a general solution of the differential equation.

(4)

Initially,

- there are no antibodies in the bloodstream of the patient
- the concentration of antibodies is estimated to be increasing at $10 \mu\text{g/ml}$ per week

- (b) Find, according to the model, the maximum concentration of antibodies in the bloodstream of the patient after the vaccination.

(8)

A second dose of the vaccine has to be given to try to ensure that it is fully effective. It is only safe to give the second dose if the concentration of antibodies in the bloodstream of the patient is less than $5 \mu\text{g/ml}$.

- (c) Determine whether, according to the model, it is safe to give the second dose of the vaccine to the patient exactly 10 weeks after the first dose.

(2)

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